

Practical 1:

Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup Problem Definition Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure README file, and GitHub repository.

Key Questions / Analysis / Interpretation

1. What is URL and parts of URL?
2. How does HTML file processed in web browser?
3. How will page navigation flow managed among all HTML pages?
4. How will GitHub commits be maintained after each practical?

Key Skills to be Addressed

Requirement analysis, sitemap design, wireframing, Git/GitHub, README writing, folder structuring Applications Web application planning and maintainable project setup Learning Outcome Students will be able to analyze requirements and prepare a structured project foundation before implementation.

Dataset/Test Data N/A; design artefacts only

Tools/Technology to be Used : VS Code, Draw.io/Figma, Git, GitHub

Total Hours Implementation: 4 hours | Total engagement: 4 hours

Post Laboratory Work

Description

Submit project functional requirements, sitemap, wireframe screenshots, GitHub repository link, and README.md.

Evaluation Strategy

Including Viva Requirement clarity, sitemap completeness, role definition, repository structure, and viva on project scope.

Advanced/Intermediate Extension

Intermediate: Create responsive wireframes in Figma.

Advanced: Prepare ER diagram and REST-style route plan for future backend modules.

Github Link: <https://github.com/MeetBhatti/WDF.git>

1. What is URL and parts of URL?

Ans. URL stands for **Uniform Resource Locator**. It is the address used to open a webpage on the Internet. URLs are like `http://.....` and `https://` where `http` means not secure, normally websites use this, and `https` means secure where data is in encrypted form, so banking and payment side websites use this `https://...`

For example: `https://www.example.com/page.html`

- `https://` → Protocol
 - `www.example.com` → Domain name
 - `/page.html` → Path/Filename
- It helps the browser find and open the required webpage

2. How does HTML file processed in web browser?

First, the browser opens the HTML file and reads its code. It understands HTML tags like `<h1>`, `<p>`, `<table>`, etc. Then, the browser creates the structure of the webpage. If CSS is connected, it applies the CSS styling to the page. Finally, the browser displays the webpage on the screen.

Example: HTML code `<h1>Hello</h1>` displays **Hello** as a heading.

3. How will page navigation flow managed among all HTML pages?

Page navigation is managed using the **`<a>` tag** and `href` attribute. The `href` contains the path of the page that we want to open. When the user clicks the link, the browser opens that HTML page.

```
<a href="profile.html">Profile</a>
```

In our StudentHub, the Home page can link to Login, Assignment, Attendance, Profile, Result, etc.

4. How will GitHub commits be maintained after each practical?

After completing each practical, we will save the changes using a **Git commit**. We can give a clear message to know what work was completed. For example:

```
git add .
```

```
git commit -m "Completed Practical 1"
```

```
git push
```

After the next practical, we can make another commit like **"Added Assignment and Result pages"**.

This helps us keep a record of our project work and changes.

About Website:

This is Student Hub Portal or Website where Student , Faculty , and Admin can visit and see content easily. Student can view and manage academic information. Faculty can Can provide assignments, notices, and academic information and Admin can manage students, faculty, and portal information.

In this website there are main we have 10 basic html webpages like
(1)Home Page

(2>Login

(3)Assignment

(4)Attendance

(5>Contact

(6)Courses

(7)Dashboard

(8)Profile

(9)Register

(10)Result

For this website some requirement are necessary according to Student, Faculty and Admin so requirement like as:

Requirement Analysis for Student Hub Portal:

In this Student Hub Portal Generally 10 webpages are there according to Student requirement , Faculty requirement and according to Admin requirement so this 10 webpages for students are like:

So from first page home page where you can see content of Student Hub Portal and from Login you can Login as Student , Login as Faculty, Login as Admin.

Reuiement :**(1)Home page:**

Content show of Website , Other Important Notice , Events Announcement , and from Login etc....

(2>Login Page:**In Login Page you can Login as:****(1)Student:**

(2)Faculty:**(3)Admin:**

By entering the correct id and password.

(3)Login as Student:

Student Dashboard:

Student can do / Show data like as:

(i)Student Fees Details:

(ii)Student Result :

(iii)Student Progress Report:

(iv)Feedback of Faculty:

(v)Attendance:

(vi)Menu

(vii)Student Profile:

(i)Student Fees details:

Total fees, remaining fees , and you paid fees:

(ii)Student Result:

First Student have to select the semester and after enter require data and show result.

(iii)Student Progress Report:

To press this button student can see progress of him or her self like in which subject good score and in which have to do some more practice etc.

(iv)Feedback of Faculty:

From this student can give feedback to their Faculty.Where some question there student have to answer it and give feedback.

(v)Attendance:

From this feature student can see him or her self attendance by lecture wise , lab wise and also particular date wise.

(vi)Menu:

In menu also other options are there like Courses details , Leave application and also like Schlorthship details etc.

(vii)Student Profile:

In profile section student can see or edit him or her personal data like as Name , Address , id , mobile no and email id and other details.

(4)Login as Faculty:

Faculty Dashboard:

Faculty can do / Show data like as:

(i)Add new Student:

(ii)Remove Student :

(iii)Student Progress Report:

(iv)Attendance of taking lecture:

(v)Student Result:

(vi)Menu:

(vii)Faculty Profile:

(i)Add new Student :

To add new student faculty has to enter the details of Student like as Name , id , contact no , Email id , Department and othe personal information.

(ii) Remove Student:

To remove the student faculty has to give reason and other information like as Name,id, Email id, Department etc.

(iii)Student Progress Report:

In this feature Faculty can see Student progress report but for this they have to give details of that student like as name and id so they can see in which subject he or she have to do more practice and in which good etc.

(iv)Attendance of taking lecture:

Here faculty can take attendance of particular lecture according to date and department which topics they cover in that lecture and take attendance.

(v)Student Result:

For this faculty have to give information of that student for which they have to see result.

(vi)Menu:

in Menu , faculty can Leave application , any notice , or courses details etc.

(vii)Faculty Profile:

In profile section Faculty can see or edit personal data like as Name , Address , id mobile no and email id ,department and Experience and other details.

(5)Login as Administator:

Admin Dashboard:

Admin can do / Show data like as:

(i)Add Faculty:

(ii)Remove Faculty:

(iii)Faculty Progress Report / Details :

(iv) Student Progress Report / Details :

(v)Add Student:

(vi)Remove Student:

(vii)Menu:

(i)Add Faculty :

To add new faculty, Admin has to enter the details like as Name , id , contact no , Email id , Department ,Experience and other specifications of Faculty and other personal information.

(ii) Remove Faculty:

To remove the faculty Admin has to give reason and other information like as Name,id, Email id, Department ,Experience , Degree ,etc.

(iii)Faculty progress report:

For this Admin can see Faculty Progress Report by entering details of faculty such as id, name and department and according to student feedback they can see progress report of faculty.

(iv)Student Progress Report:

In this feature Admin can see Student progress report but for this they have to give details of that student like as name and id so they can see in which subject he or she have to do more practice and in which good etc.

(v) Add Student :

For this Admin can also add student so for this Admin has to give information of that student like as Name , id , Contact Number, marks , Department etc.

(vi)Remove Student:

Admin has to give reason and some basic details of that student so can remove.

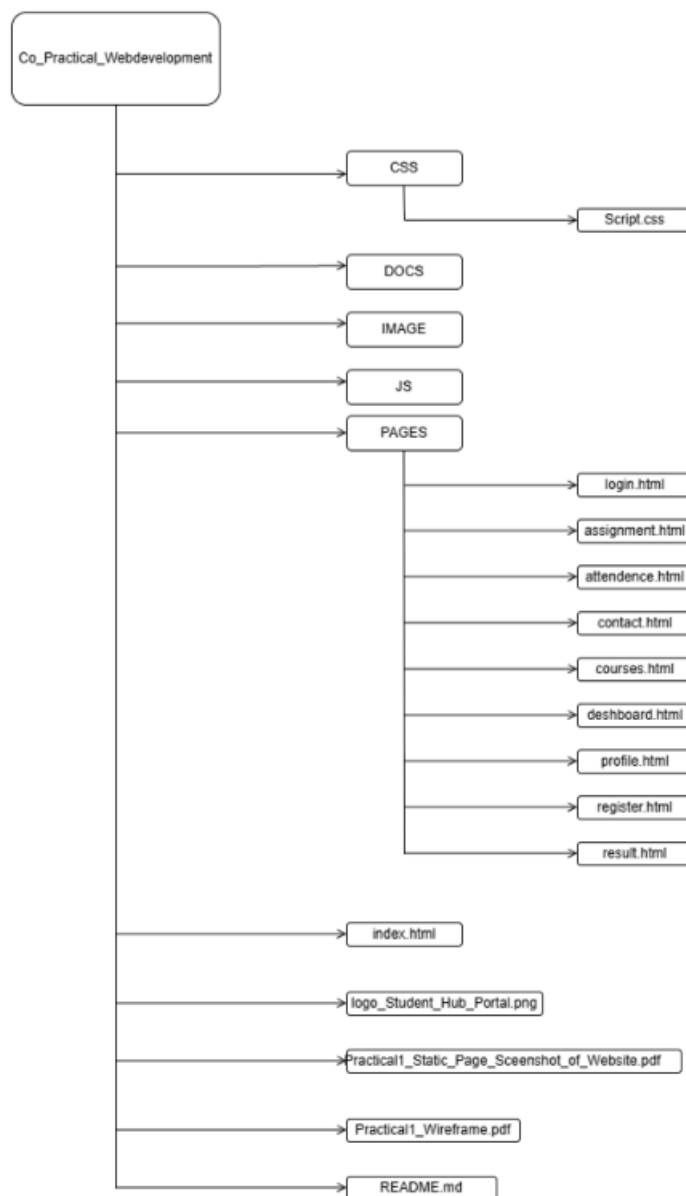
(vii)Menu:

In menu Admin can Announce , Give Notice to Faculty , Settings some change can make , show leave application of faculty and student.

Sitemap:

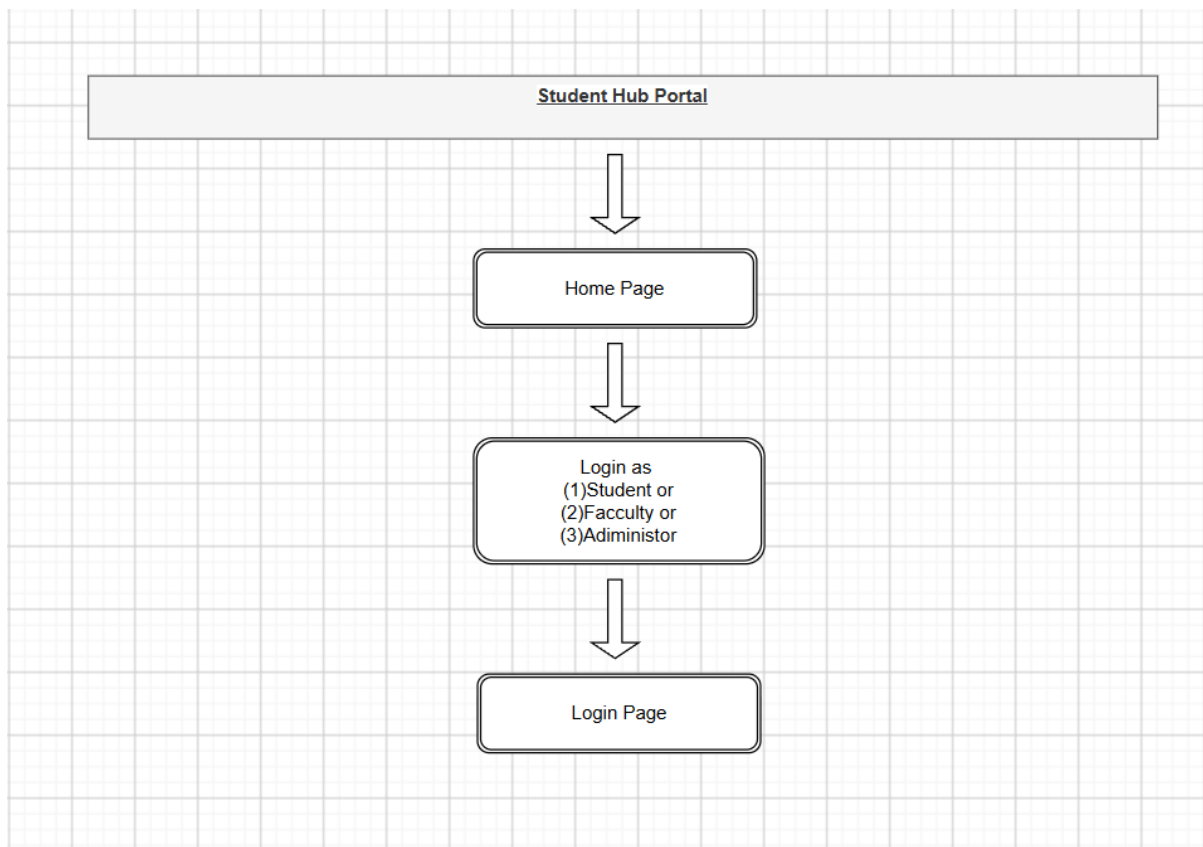
A sitemap shows the structure of the StudentHub Portal and how different pages are connected with each other.

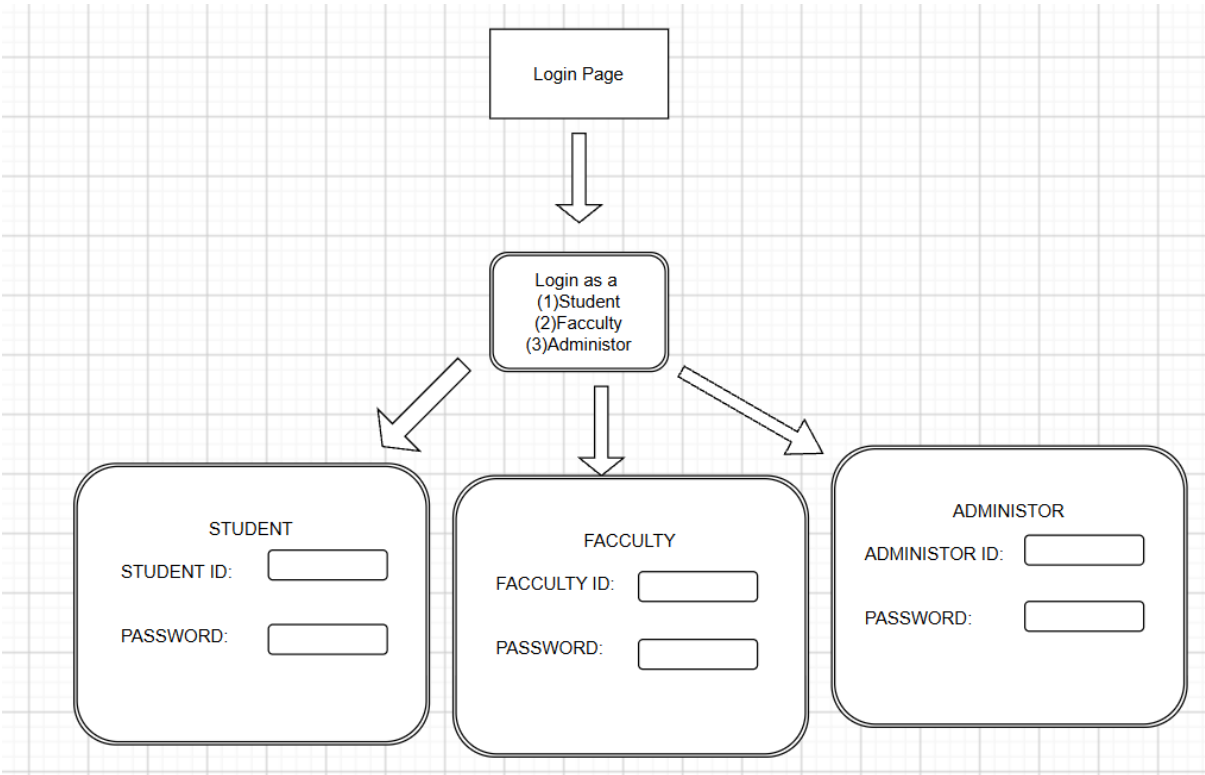
Sitemap Of Student Hub Portal



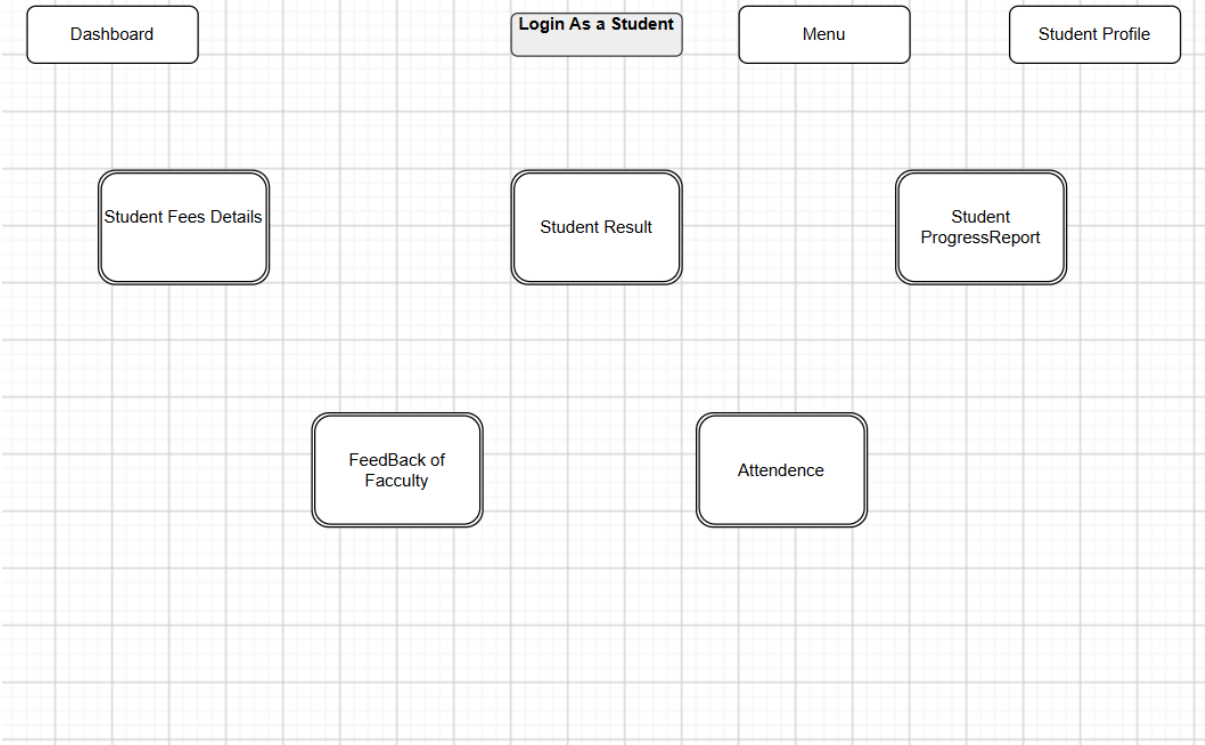
WireFrame:

A wireframe is a basic design of a webpage. It shows where important elements such as the navigation bar, headings, buttons, forms, and content will be placed and also in another words wireframe means blue print of webpage it uses for layout how our webpage will look after .

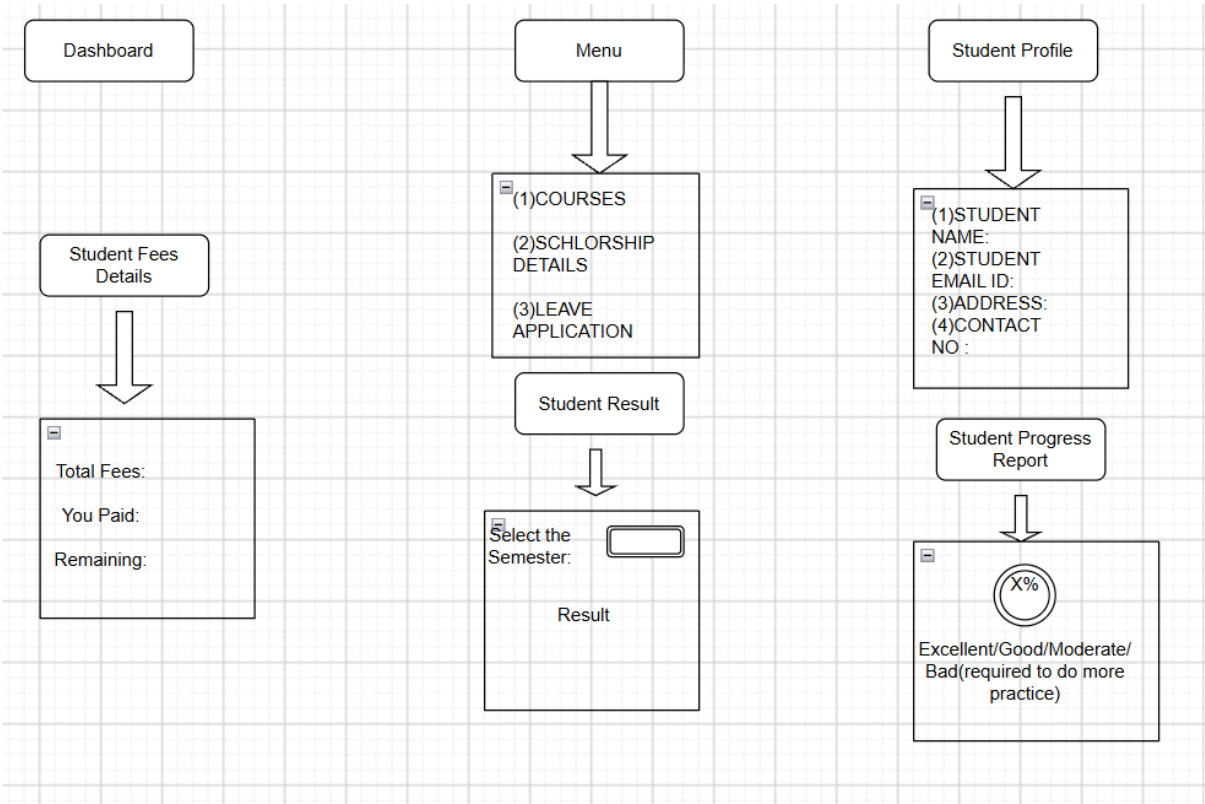
Wireframe of Home Page:**Wireframe of Login Page:**



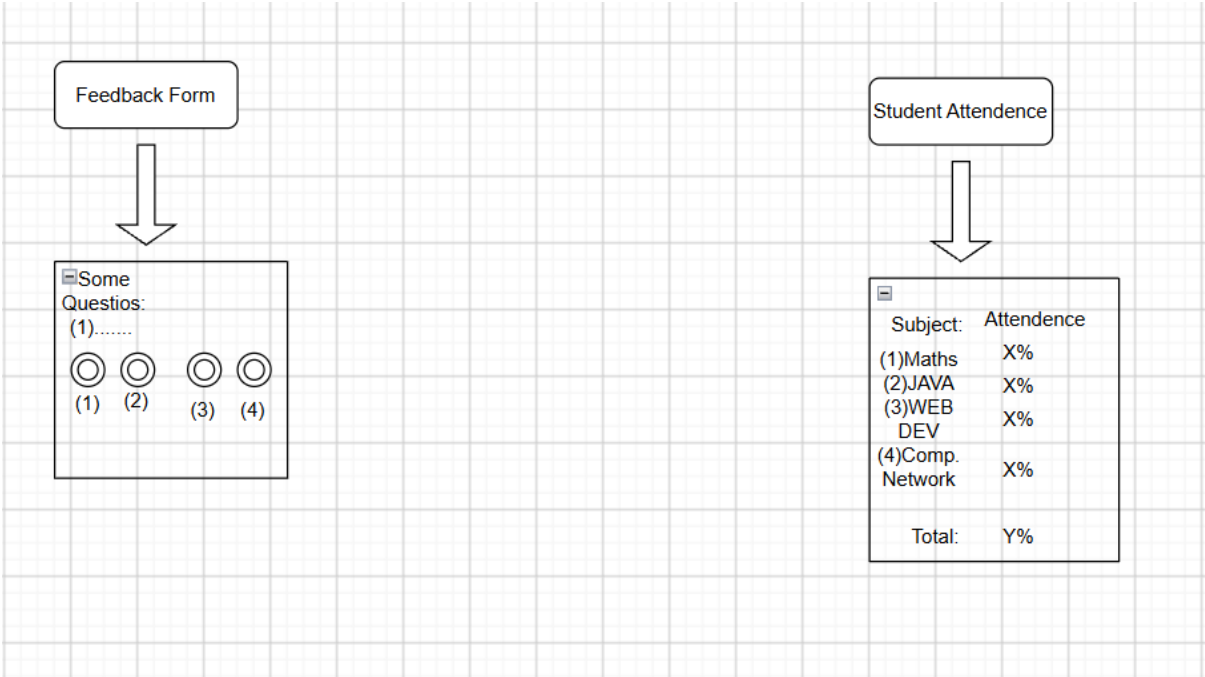
Wireframe of Login as Student (Page open for student requirement):



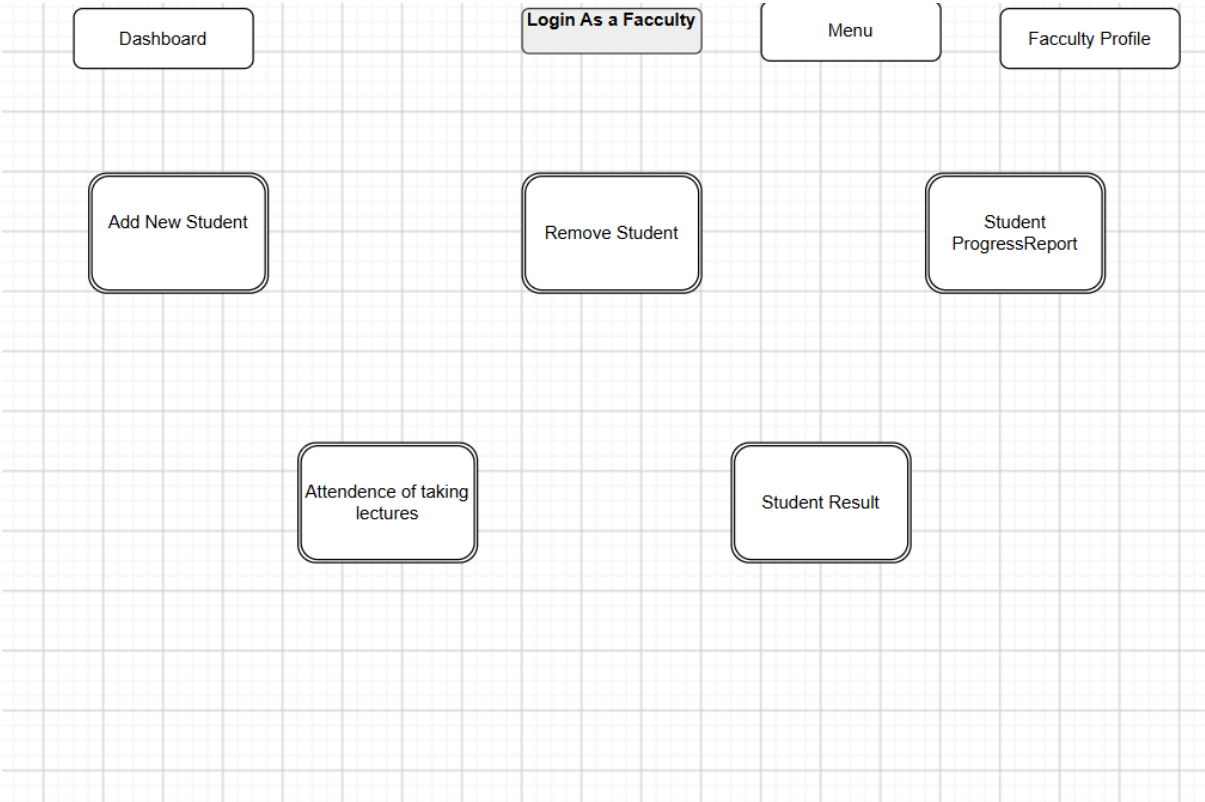
Wireframe of Student Dashboard:



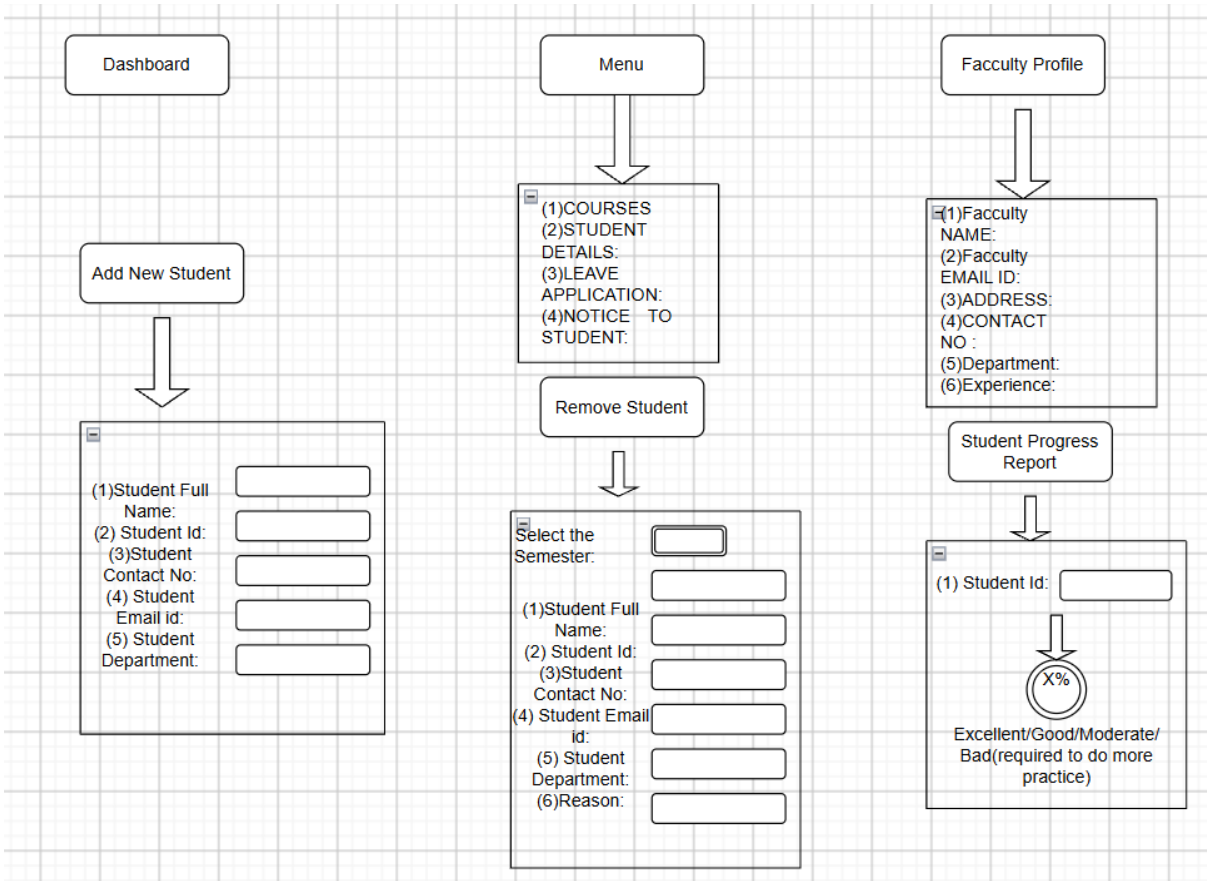
Wireframe of Student Dashboard:



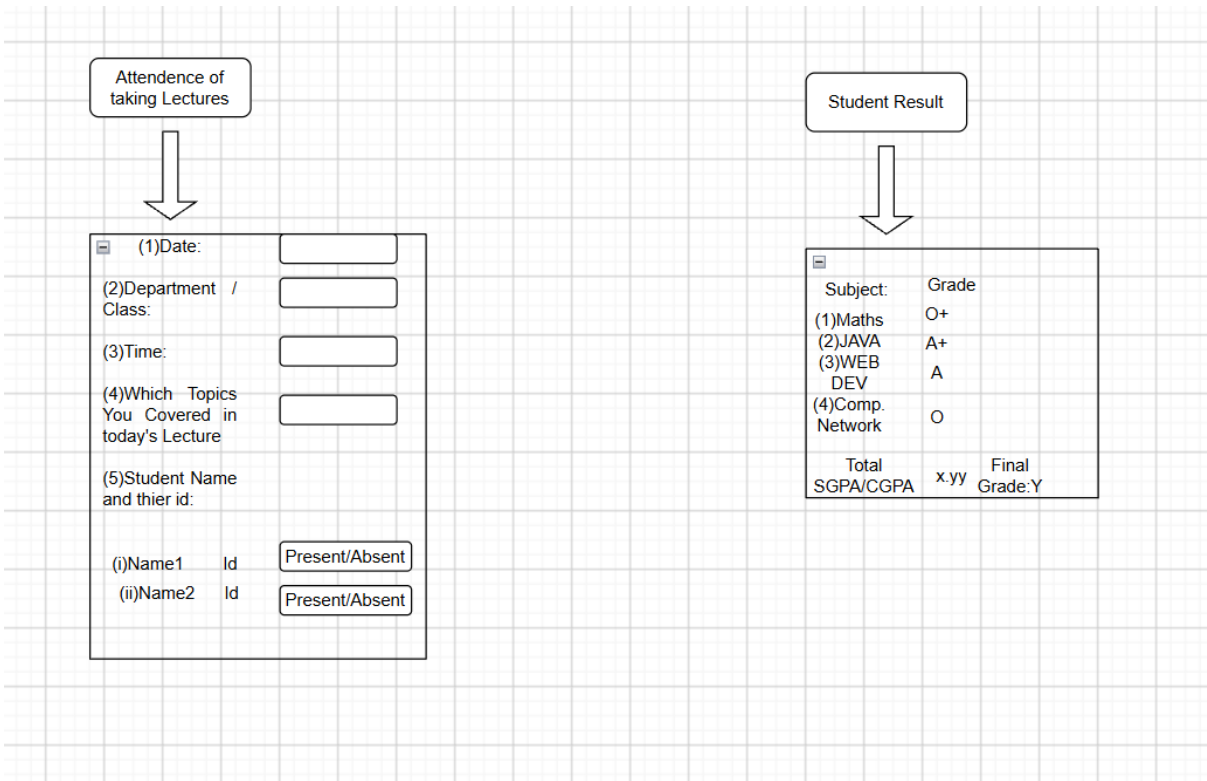
Wireframe of Login as Faculty (Page open for Faculty requirement):



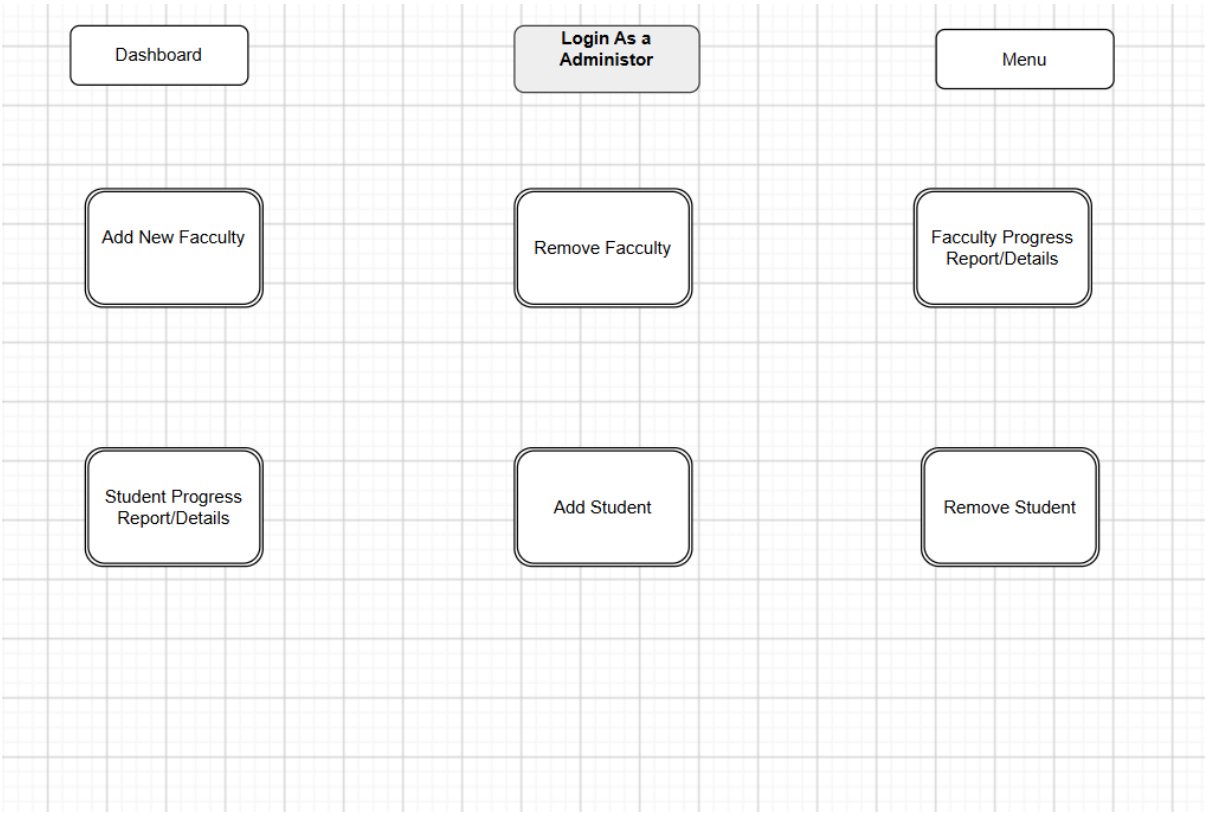
Wireframe of Faculty Dashboard:



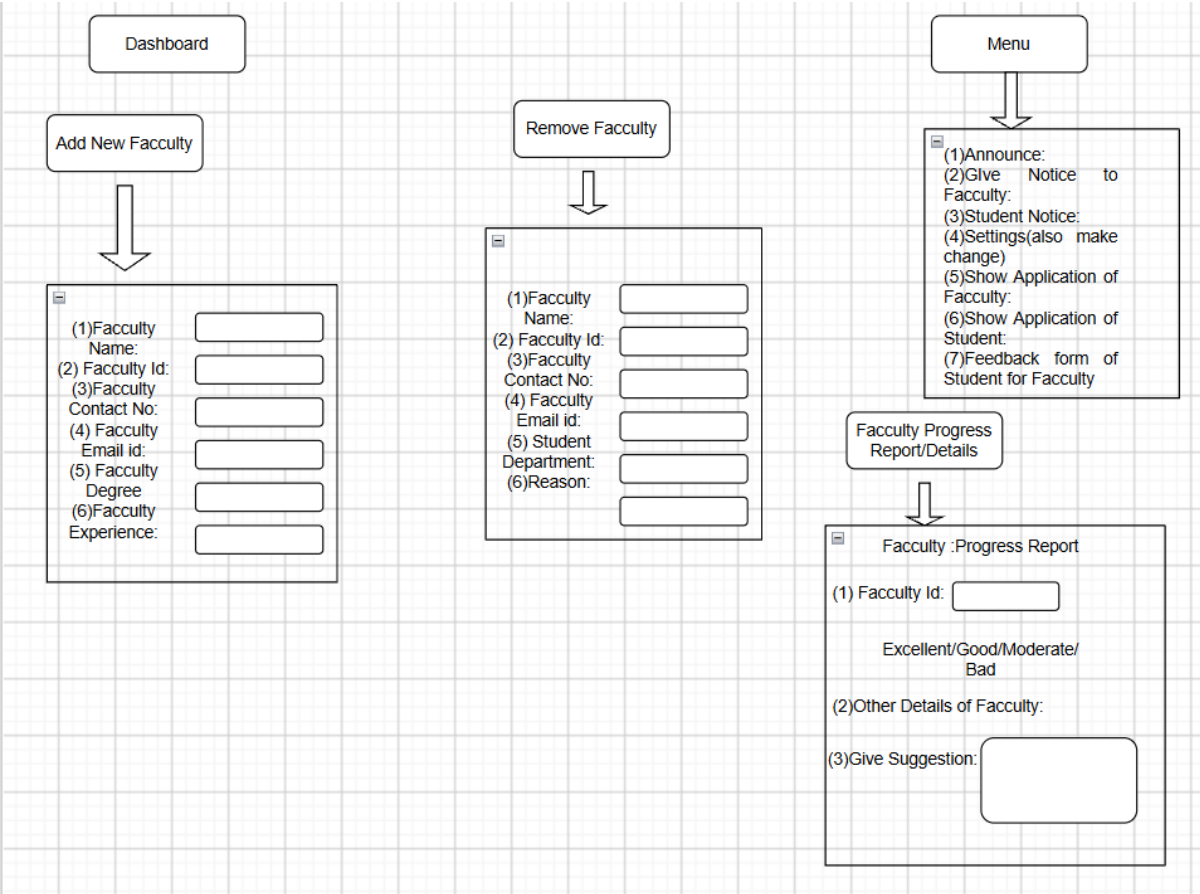
Wireframe of Faculty Dashboard:



Wireframe of Login as Admin:



Wireframe of Admin Dashboard:



Wireframe of Admin Dashboard:

